

Movement

Fitts's Law

Fitts law

- Fitts aimed to find the capacity of human movement — how many repetitive movements could be performed in a given time interval. He did several experiments involving such repetitive movements and as a result he came up with Fitts's law, one of the most famous laws of human-computer interaction.

Fitt's law

- Fitts's law gives us the relationship between the time it takes a **pointer** (such as a mouse cursor, a human finger, or a hand) to move to a particular **target** (e.g., physical or digital button, a physical object) in order to interact with it in some way (e.g., by clicking or tapping it, grasping it, etc.):

$$T = a + b \log_2 \frac{2D}{w}$$

Fitt's law

- where D is the distance to the target, w is the width of the target (technically, measured along the axis of movement), and a and b are constants that vary depending on the type of pointer (e.g., mouse, finger, etc.).

Fitts law

- The quantity is sometimes called an index of difficulty and is measured in bits.

$$ID = \log_2 \frac{2D}{w}$$

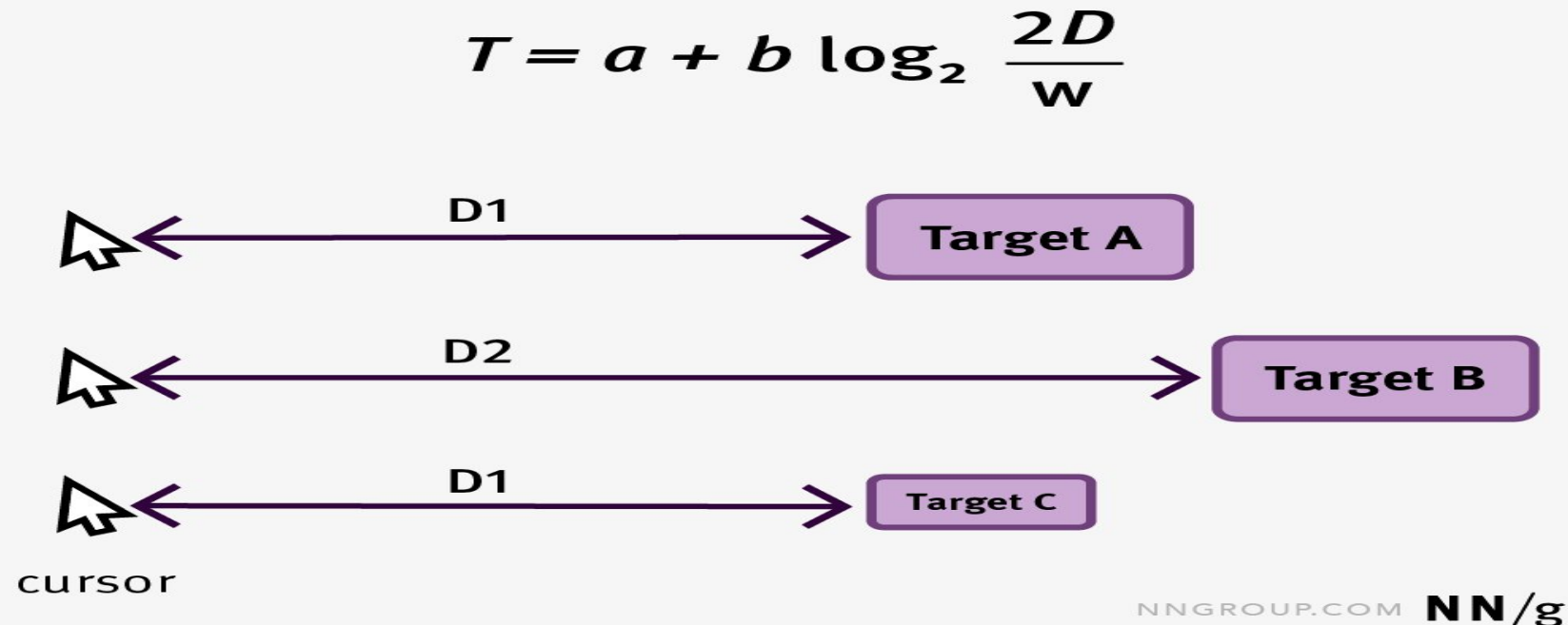
- An alternative formulation of Fitts's law is then:

$$T = a + b ID$$

Fitts law

- If this mathematical equation seems obscure to some of you, its gist is captured by the following two statements below:
- **The bigger the distance to the target, the longer it will take for the pointer to move to it.** In other words, closer targets are faster to acquire.
- **The larger the target, the shorter the movement time to it.** In other words, bigger targets are better.

Fitts' s law says that the time to reach Target A is shorter than the time to reach any of the other targets. Although Targets A and B have the same size, the distance from cursor to A (D1) is shorter than the distance to B (D2), so movement to A will be faster. Target C is placed at the same distance (D1) from the cursor as Target A, but it's smaller, so it will take longer to move the cursor to it than to A.



Fitts law

- The equation includes a logarithm — meaning that the time doesn't grow linearly with the parameters, but at a slower pace. So, if something is twice as far away, it takes longer to reach it, but not twice as long. That is because movement first accelerates and then decelerates (as described in the next section), meaning that you get to move the pointer at a higher speed during part of the travel time when the distance is longer, and then more slowly as you get closer.

Two-Component Model

The model said that any goal-directed movement (e.g., performed to acquire a target as opposed to, say, tapping in the rhythm of music) can be decomposed into two components (hence the name):

- an **initial movement**, rapid and relatively coarse, which is intended to move the pointer in the direction of the target
- a **final movement**, which is slower and intended to secure accuracy (for example, ensure that the target is acquired correctly)

Fitts law

- The first phase is largely driven by the distance to the target — the movement will aim to take the pointer in the direction of the target as fast as possible. **The second movement phase is, however, driven mostly by size of the target** — the pointer needs to **slow down in order to not overshoot and click beyond the target**. The smaller the target, the more time tends to be spent in this second phase.

Applications to UX

- The implications of Fitts's law can be grouped into 2 distinct categories, corresponding to the two variables that affect the movement time: target size and distance to target.

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Optimizing Target Size

Bigger Is Better

- The most obvious implication of Fitts's law is: make targets big. Fitts's law clearly says that people will be faster to click, tap, or hover on bigger targets. Not only that, but error rates go down as target sizes increases. Guidelines for target size are usually created by determining the rate of error for various target sizes and identifying the size at which the error rate levels off.

Optimizing Target Size

Icons Plus Labels

- We've said many times that icons need labels. Not only does a text label reduce icon ambiguity and makes it easy to understand, but it also improves movement time to that particular target. Why? It's simple: any target made up of both an icon and a label will be greater than just an icon and, therefore, according to Fitts's law, will be easier to acquire. (This, of course, assumes that you do make the entire area clickable and avoid the design mistake of only making the icon itself the active target.)



Fitts law

- Read on **Infinite Targets Along Screen Edges – comparing Mac-OS and windows**

Fitts law

- **Don't Crowd Targets**
- If you place targets too close to each other, there is a risk that people will accidentally overshoot and accidentally trigger the wrong target. Note that this is especially likely to happen if the targets are small. **What is a dead space?**
- **Padding Is Not Enough**
- Sometimes designers will pad certain targets, thus increasing the effective target area. However, remember that, even though padding will help prevent some overshooting errors, if people do not realize that a target is padded, they may still cautiously slow down more than necessary as they get closer to the target, in order to make sure they will not go beyond the target area.
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GOMS and Keystroke-level model

GOMS

Learning Goals

- Know about the GOMS family
- Be able to use GOMS for simple tasks

GOMS

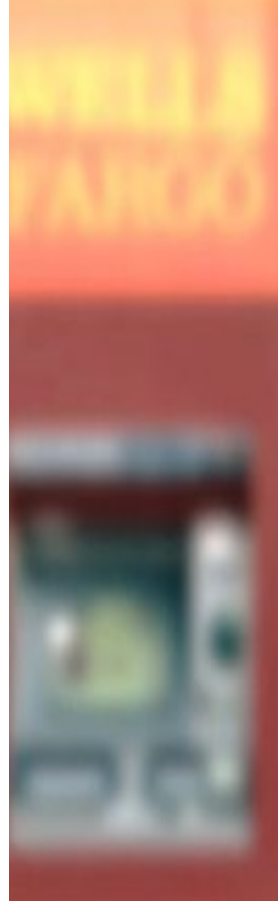
- GOAL: GET-MONEY
- GOAL: USE-CASH-MACHINE
 - INSERT-CARD
 - ENTER-PIN
 - SELECT-GET-CASH
 - ENTER-AMOUNT
 - COLLECT-MONEY

outer goal satisfied



- GOAL: GET-MONEY
- GOAL: USE-CASH-MACHINE
 - INSERT-CARD
 - ENTER-PIN
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 - ENTER-AMOUNT
 - COLLECT-CARD
 - COLLECT-MONEY

outer goal satisfied

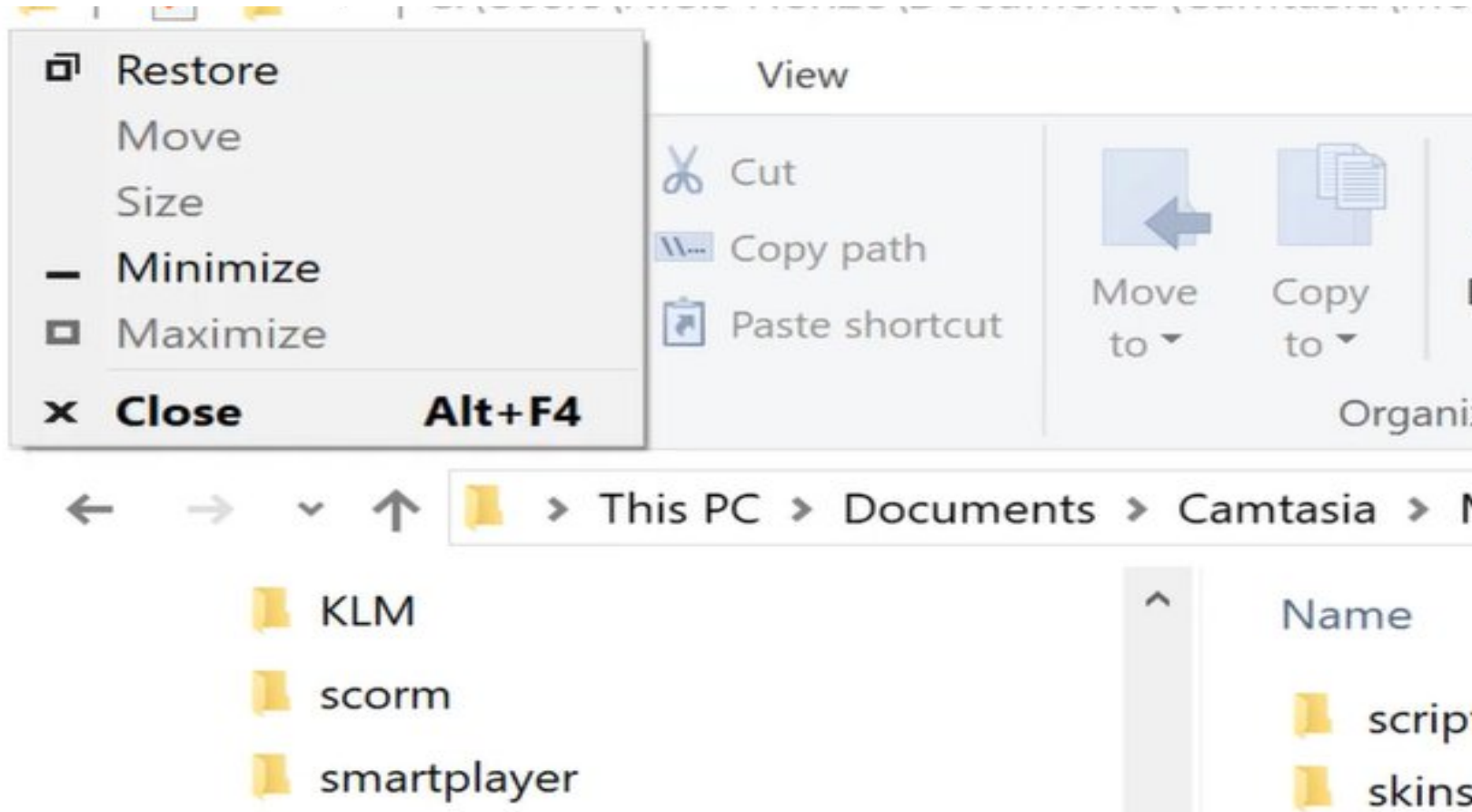


GOMS model

The GOMS Model

- **Goals**
 - (Verbal) description of what a user wants to accomplish
 - Various levels of complexity possible
- **Operators**
 - Possible actions in the system
 - Various levels of abstraction possible (sub-goals / ... / keystrokes)
- **Methods**
 - Sequences of operators that achieve a goal
- **Selection rules**
 - Rules that define when a user employs which method (among alternatives)

For example – closing a widow



GOMS

GOAL: CLOSE-WINDOW

[select

GOAL: USE-MENU-METHOD

MOVE-MOUSE-TO-FILE-MENU

PULL-DOWN-FILE-MENU

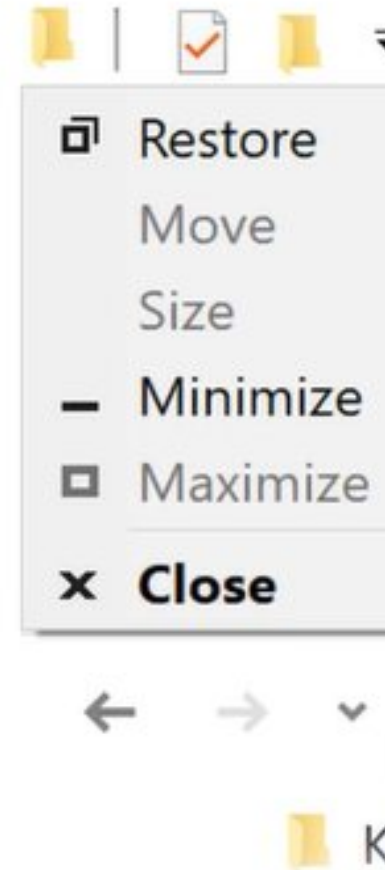
CLICK-OVER-CLOSE-OPTION

GOAL: USE-ALT-F4-METHOD

HOLD-ALT-KEY

PRESS-F4-KEY]

VERIFY-CLOSE



GOMS – Characteristics

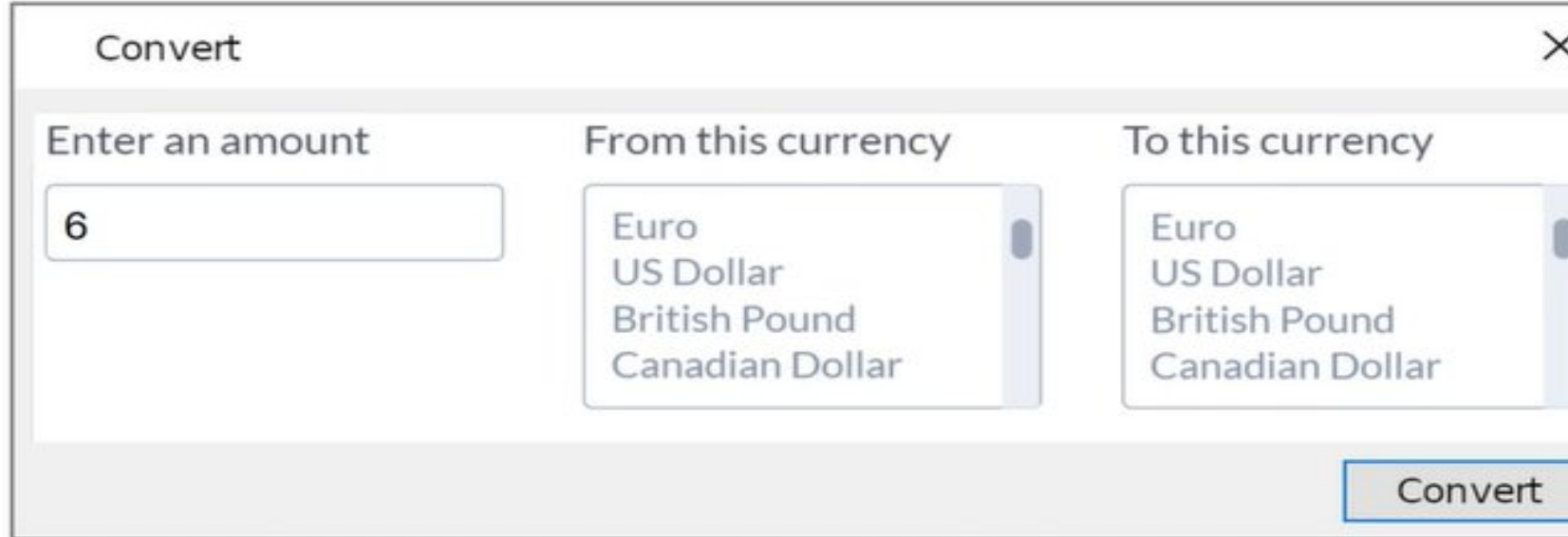
- Can be used to model complex tasks
- Cannot predict completion times
- But the simpler KLM can
- Predictions
 - More operators, longer completion
 - Deep depth of goal structure → high short term-memory load
 - Users stop when goals are satisfied

Keystroke-Level Model

Learning Goals

- Know what KLM stands for
- Know the KLM operators
- Being able to predict how long tasks take using KLM

Keystroke-Level Model



The screenshot shows a web-based currency converter interface. At the top, there is a title bar with the word "Convert" on the left and a close button (an 'X' icon) on the right. Below the title bar, the interface is divided into three main sections: "Enter an amount", "From this currency", and "To this currency". In the "Enter an amount" section, there is a text input field containing the number "6". The "From this currency" section features a dropdown menu with four options: "Euro", "US Dollar", "British Pound", and "Canadian Dollar". The "To this currency" section also has a dropdown menu with the same four options. At the bottom right of the interface, there is a button labeled "Convert".

- Task: Convert 12 Euro in US Dollar
- One hand on the mouse, nothing selected
- What do we need to know?

Keystroke-Level Model

- Task: Convert 12 Euro in US Dollar
- One hand on the mouse, nothing selected

Convert

Enter an amount

12

From this currency

Euro
US Dollar
British Pound
Canadian Dollar

To this currency

Euro
US Dollar
British Pound
Canadian Dollar

Convert

select text field
delete value
enter value
select Euro
select Dollar
select Convert

Keystroke-Level Model (KLM)

- Simplified version of the "**G**oals, **O**perators, **M**ethods, and **S**elections rules" (GOMS) Model
- KLM predicts how much time it takes to execute a task

KLM

- Simplified version of the "**G**oals, **O**perators, **M**ethods, and **S**elections rules" (GOMS) Model
- KLM predicts how much time it takes to execute a task
- Execution of a task is decomposed into primitive operators
 - Physical motor operators
 - Pressing a button, pointing, drawing a line, ...
 - Mental operator
 - Preparing for a physical action
 - System response operator
 - User waits for the system to do something

List of KLM operators

Operator	Description	Associated Time
K	Keystroke, typing one letter, number, etc. or function key such as 'CTRL' or 'SHIFT'	Expert typist (90 wpm): 0.12s Averaged skilled typist (55 wpm): 0.20s Average non-secretarial typist (40 wpm): 0.28s Worst typist (unfamiliar with keyboard): 1.2s
H	'Homing', moving the hand between mouse and keyboard	0.4s
B/BB	Pressing (B) or clicking (BB) a button	0.1s / 2*0.1s
P	Pointing with a mouse to a target	0.8s to 1.5s with an average of 1.1s Can also use Fitts' Law
$D(n_D, l_D)$	Drawing n_D straight line segments of length l_D	$0.9s * n_D + 0.16 * l_D$
M	Subsumed time for mental acts; sometimes used as 'look-at'	1.35s
R(t)	System response time, time during which the user cannot act	Dependent on the system

KLM

- Task: Convert 12 Euro in US Dollar
- One hand on the mouse, nothing selected

Convert

Enter an amount

1

From this currency

- Euro - EUR
- United States Dollars - USD
- United Kingdom Pounds - GBP
- Canada Dollars - CAD

To this currency

- Euro - EUR
- United States Dollars - USD
- United Kingdom Pounds - GBP
- Canada Dollars - CAD

Convert

select text field

delete value

enter value

select Euro

select Dollar

select Convert

KLM

- Task: Convert 12 Euro in US Dollar
- One hand on the mouse, nothing selected

Convert

Enter an amount

1

From this currency

Euro - EUR
United States Dollars - USD
United Kingdom Pounds - GBP
Canada Dollars - CAD

To this currency

Euro - EUR
United States Dollars - USD
United Kingdom Pounds - GBP
Canada Dollars - CAD

Convert

select text field

P, BB

delete value

H, K

enter value

M, K, K

select Euro

H, M, P, BB

select Dollar

M, P, BB

select Convert

P, BB

KLM

select text field	P, BB
delete value	H, K
enter value	M, K, K
select Euro	H, M, P, BB
select Dollar	M, P, BB
select Convert	P, BB

Operator Times:

$P \approx 1.1s$	$B = 0.1s$	$H = 0.4s$
$M = 1.35s$	$K = 0.28s$	

$4 * P =$	4.40s
$8 * B =$	0.80s
$2 * H =$	0.80s
$3 * M =$	4.05s
$3 * K =$	0.84s
Total =	10,89s

Real comparison

Version 1

- ☐ Mail
- ☐ Text
- ☐ Web
- ☐ Photo

Go

Version 2

[Mail](#)

[Text](#)

[Web](#)

[Photo](#)

Version 3

Mail



Go

Version 4

|

Go

KLM

Version 1

☐ Mail

☐ Text

☐ Web

☐ Photo

Go

Version 2

[Mail](#)

[Text](#)

[Web](#)

[Photo](#)

Hand on mouse, nothing selected,
go to photo:

- Which is the fastest interface?
- Which is the slowest?

Version 3

Mail ▼

Go

Version 4

|

Go

Conclusion

- The Keystroke-Level Model predicts task completion time for simple dialogs
- Assumes a trained average user
- Especially useful to compare alternatives
- Using KLM by hand can become lengthy and complex
- KLM is not useful for tasks that require reasoning

Read on :

- theories - page 82- 91(GOMS)
- chapter 9 (9.3.5 Fitt's law)

Reference book